CPSC 3660 — Practice Final Exam
Full Solutions & Explanations

Predictive practice based on the professor's construction signature

University of Lethbridge; 180 minutes; 120 marks

Each question below shows: the question, a **mirrors** line noting which past assignment/midterm pattern it echoes, the **Your Answer** block (what you would write on the exam), and an **Understanding** block explaining every line of reasoning.

About this solutions guide. Each question is mapped back to the professor's recycled-schema patterns (Library 85%, Employee-Works-Company 80%, Bank 70%), so once you finish this practice, you recognize the real exam's questions as variations on patterns you've already seen.
Online companion: gbasran.github.io/cpsc3660-course-guide
Blank practice exam: Practice Final PDF (no solutions)
Ground-up study guide: Complete Study Guide PDF (41 pages)
Professor's Signature: forensic analysis of exam construction patterns

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Question 1 (2 marks) — DML vs DDL; Basic

Question. Explain the difference between DML and DDL. Give one example of each.

This mirrors: *Midterm Q1, A1 Q1 (recurring opener).*

✓ Your Answer

DDL (Data Definition Language) modifies the database *schema*. Examples: `CREATE`, `ALTER`, `DROP`, `TRUNCATE`.

DML (Data Manipulation Language) modifies the *data* inside existing tables. Examples: `INSERT`, `UPDATE`, `DELETE`, `SELECT`.

Understanding

DDL answers "what is the shape of the container?" and DML answers "what is inside the container?". The prof always wants one word on the distinction (schema vs data) plus at least one concrete keyword on each side.

Question 2 (2 marks) — Key Types; Basic

Question. Difference between superkey, candidate key, primary key. Can a relation have more than one candidate key?

This mirrors: *Midterm Q2, A1 Q3.*

✓ Your Answer

Superkey: any set of attributes whose values uniquely identify each tuple of a relation. **Candidate key**: a *minimal* superkey (no proper subset is also a superkey). **Primary key**: the candidate key the designer *chooses* as the main identifier.

Yes, a relation can have several candidate keys. For example in `employee(SIN, emp_id, name)`, both {SIN} and {emp_id} are candidate keys, and the designer picks one as the primary key.

Understanding

Hierarchy is *superkey* $\supseteq$ *candidate key* $\supseteq$ *primary key* (with size 1 at the right). Every candidate key is a superkey; not every superkey is a candidate key.

Question 3 (2 marks) — Strong vs Weak Entity; Basic

Question. Distinguish strong and weak entity sets. Give one example of each.

This mirrors: *A3 Q5, A3 Q7 (loan/payment case).*

✓ Your Answer

A **strong entity** has its own primary key and exists independently. Example: `loan(loan_number, amount)` — `loan_number` is a PK; loans exist on their own.

A **weak entity** has no primary key of its own; its identifier is the PK of the *owner* plus a *discriminator* unique within that owner. Example: `payment(payment_no, date, amt)` depends on loan. Payment #1 of loan L-15 is different from payment #1 of loan L-17; only (`loan_number`, `payment_no`) together identify a payment.

Understanding

Visually: strong entity = single rectangle, weak entity = double rectangle; the identifying relationship is a double diamond; the weak entity has total participation (double line) in that relationship.

Question 4 (2 marks) — RA Intersection via Basic Ops; Basic

Question. Using only $\sigma, \pi, \times, \cup, -$, express $r \cap s$.

This mirrors: *A1 Q4, Midterm Q6.*

✓ Your Answer

$$r \cap s = r - (r - s).$$

Understanding

Intuition: $r - s$ is "everything in r but not in s". Subtracting that from r leaves behind exactly the tuples r shares with s, which is the intersection. The operator $\cap$ is therefore *derived*, not primitive. The six primitives are $\sigma, \pi, \times, \cup, -, \rho$ (rename).

Question 5 (2 marks) — BCNF Definition; Basic

Question. Define BCNF. Why do we aim for it?

This mirrors: *A4 Q2, expanded for final scope.*

✓ Your Answer

A relation R is in **Boyce–Codd Normal Form (BCNF)** with respect to FD set F iff for every non-trivial FD $\alpha \to \beta$ in F^+ that holds on R, α is a **superkey** of R.

We aim for BCNF because any FD whose LHS is not a superkey causes redundancy: the RHS value is forced to repeat across every tuple sharing the LHS. This redundancy creates update, insertion, and deletion anomalies. BCNF eliminates those anomalies structurally by permitting only FDs whose LHS uniquely determines each tuple.

Understanding

Prof's memorized one-liner: "For every non-trivial FD, the LHS must be a superkey." Trivial means $\beta \subseteq \alpha$; those are exempt.

Question 6 (2 marks) — FD Definition + Counterexample; Basic

Question. What does $\alpha \to \beta$ mean? Give a counterexample showing $\alpha \to \beta \not\Rightarrow \beta \to \alpha$.

This mirrors: *A4 Q1, expanded for final scope.*

✓ Your Answer

$\alpha \to \beta$ means: for any two tuples t_1, t_2 of R, if $t_1[\alpha] = t_2[\alpha]$ then $t_1[\beta] = t_2[\beta]$. Intuitively, α *functionally determines* β; the value of β is a function of the value of α.

Counterexample. In `student(student_id, student_name)`, `student_id` $\to$ `student_name` holds because IDs are unique. But `student_name` $\to$ `student_id` fails: two different students can share the same name ("John Smith" could map to ID 1001 or ID 2003), so `student_name` does not determine `student_id`.

Understanding

FDs are not symmetric. The notation $\alpha \to \beta$ is directional; drawing the reverse arrow requires an independent check.

Question 7 (10 marks) — SQL; Hospital schema; Medium

Schema. `patient(patient_id, patient_name, patient_city, age)`; `doctor(doctor_id, doctor_name, specialty)`; `appointment(patient_id, doctor_id, appt_date, fee)`.

This mirrors: *A2 Q7 + Midterm Q6–Q7. Structurally analogous to the Employee–Works–Company trio; a three-table join with one filter.*

(a) (3 marks) Patients who saw a cardiologist.

✓ **Your Answer**

```sql
select distinct p.patient_name
from patient as p, doctor as d, appointment as a
where p.patient_id = a.patient_id
  and a.doctor_id = d.doctor_id
  and d.specialty = 'Cardiology';
```

Understanding

Classic three-table join. Join `patient` to `appointment` on `patient_id`; join `appointment` to `doctor` on `doctor_id`; filter specialty. `distinct` removes duplicates from patients who have multiple cardiology appointments.

(b) (3 marks) Average fee per specialty, only where avg ¿ 200.

✓ **Your Answer**

```sql
select d.specialty, avg(a.fee) as avg_fee
from doctor as d, appointment as a
where d.doctor_id = a.doctor_id
group by d.specialty
having avg(a.fee) > 200;
```

Understanding

Prof's signature "GROUP BY with threshold" question (see Professor's Signature p.12 cheat-sheet). The aggregate `avg(a.fee)` must go in `having`, not `where`, because `where` filters rows *before* grouping while `having` filters groups *after*.

(c) (4 marks) Patients with more than the average number of appointments per patient.

✓ **Your Answer**

```sql
select p.patient_name
from patient as p, appointment as a
where p.patient_id = a.patient_id
group by p.patient_id, p.patient_name
having count(*) > (
    select count(*) * 1.0 / count(distinct patient_id)
    from appointment
);
```

Understanding

Two aggregates at play. Outer: count this patient's appointments via `group by` + `count(*)`. Inner: overall average appointments per patient is $\frac{\text{total appointments}}{\text{distinct patients}}$. The `* 1.0` forces floating-point division (otherwise integer division truncates). Compare the per-patient count to the global average with `having`.

Question 8 (12 marks) — SQL; Airline schema; Medium-Hard

Schema. `staff(staff_id, staff_name, staff_city)`; `airport(airport_code, airport_city)`; `route(staff_id, airport_code, flights_assigned)`; `supervises(staff_id, supervisor_id)`.

This mirrors: *A2 Q7 + A3 Q3 + Midterm Q9. The supervises relation mirrors the manages relation; part (c) mirrors the "¿ ALL" pattern.*

(a) (3 marks) Staff living in the same city as their supervisor.

✓ **Your Answer**

```sql
select s1.staff_name
from staff as s1, supervises as sp, staff as s2
where s1.staff_id = sp.staff_id
  and sp.supervisor_id = s2.staff_id
  and s1.staff_city = s2.staff_city;
```

Understanding

Self-join trick: alias `staff` twice. `s1` is the subordinate, `s2` is the supervisor; `supervises` sits between them to connect `staff_id` → `supervisor_id`. Then compare cities.

(b) (4 marks) View `busy_staff` with total flights ≥ 50.

✓ **Your Answer**

```sql
create view busy_staff as
select s.staff_id, s.staff_name, sum(r.flights_assigned) as
   total_flights
from staff as s, route as r
where s.staff_id = r.staff_id
group by s.staff_id, s.staff_name
having sum(r.flights_assigned) >= 50;
```

Understanding

`create view name as` wraps any `select`. Sum the flights across all airports (so group by the staff key), then restrict to totals ≥ 50 via `having`. Non-aggregated columns in `select` must

appear in `group by` (hence `staff_id, staff_name` both listed).

(c) (5 marks) Staff whose flights at every airport exceed those of staff 100.

✓ **Your Answer**

```
select s.staff_name
from staff as s
where s.staff_id <> 100
  and not exists (
      select r100.airport_code
      from route as r100
      where r100.staff_id = 100
        and r100.flights_assigned >= all (
            select r.flights_assigned
            from route as r
            where r.staff_id = s.staff_id
              and r.airport_code = r100.airport_code
        )
  );
```

Equivalent formulation using `ALL`:

```
select s.staff_name
from staff as s
where s.staff_id <> 100
  and not exists (
      select *
      from route as r100
      where r100.staff_id = 100
        and not exists (
            select *
            from route as r
            where r.staff_id = s.staff_id
              and r.airport_code = r100.airport_code
              and r.flights_assigned > r100.flights_assigned
        )
  );
```

Understanding

"For every airport where 100 has an assignment, this staff member has strictly more" is a universal-quantification pattern. Translate "for every x in B, $P(x)$" into "NOT EXISTS x in B where NOT $P(x)$". Here B = airports where 100 is assigned, and $P(x)$ = "this staff has strictly more flights at x than 100 does". The double-nested NOT EXISTS is the textbook rendering.

Question 9 (10 marks) — SQL Division; Supplier schema; Hard

Schema. `supplier(supplier_id, supplier_name, supplier_city)`; `department(dept_id, dept_name, dept_city)`; `supplies(supplier_id, dept_id, item_count)`.

This mirrors: *A3 Q3(b) ("companies in every city where IBM is") and the division-via-EXCEPT pattern.*

(a) (4 marks) Suppliers serving at least one Calgary department.

✓ **Your Answer**

```sql
select distinct s.supplier_name
from supplier as s, supplies as sup, department as d
where s.supplier_id = sup.supplier_id
  and sup.dept_id = d.dept_id
  and d.dept_city = 'Calgary';
```

Understanding

Existential quantifier = three-table join + city filter. Dedupe with `distinct` since one supplier can serve multiple Calgary departments.

(b) (6 marks) Suppliers serving every department.

✓ **Your Answer**

```sql
select s.supplier_name
from supplier as s
where not exists (
    (select dept_id from department)
    except
    (select sup.dept_id
     from supplies as sup
     where sup.supplier_id = s.supplier_id)
);
```

Understanding

The canonical "A contains B ≡ `NOT EXISTS` (B `EXCEPT` A)" template. Here B = set of all department ids; A = set of department ids this supplier serves. If $B \setminus A$ is empty, the supplier covers every department.

Question 10 (12 marks) — RA + DDL; Researcher–Papers; Medium-Hard

Schema. `researcher(rid, name, affiliation, years_active)`; `paper(pid, title, authors, venue)`; `cites(rid, pid, citation_date)`.

This mirrors: *A1 Q7 + A2 Q1 + Midterm Q8 (Library-analog).*

(a) (2 marks) Researchers who cited at least one Nature paper.

✓ **Your Answer**

$$\pi_{\text{name}}(\text{researcher} \bowtie \text{cites} \bowtie \sigma_{\text{venue='Nature'}}(\text{paper}))\,.$$

Understanding

Natural joins line up on the shared attributes (`rid` between researcher and cites; `pid` between cites and paper). Filter papers to Nature before joining to shrink the intermediate result. Project `name` at the end.

(b) (3 marks) Researchers who cited every Nature paper.

✓ **Your Answer**

$$T = \pi_{\text{rid,pid}}(\text{cites}) \div \pi_{\text{pid}}(\sigma_{\text{venue='Nature'}}(\text{paper}))\,,$$
$$\text{result} = \pi_{\text{name}}(\text{researcher} \bowtie T).$$

Understanding

Division $R \div S$ returns $\{t : \forall s \in S,\ (t, s) \in R\}$. Dividend: (`rid`, `pid`) citation pairs. Divisor: all Nature `pid`s. Quotient: the `rid`s that appear paired with *every* Nature `pid`. Join back with researcher to translate rids into names.

(c) (4 marks) Per venue, researchers who cited more than five papers from that venue.

✓ **Your Answer**

Let $J = \text{cites} \bowtie \text{paper}$. Then

$$G = {}_{\text{venue, rid}}\gamma_{\text{count-distinct(pid)}\to\text{cnt}}(J),$$
$$\text{result} = \pi_{\text{name,venue}}(\text{researcher} \bowtie \sigma_{\text{cnt}>5}(G))\,.$$

Understanding

The grouping operator $\gamma_G f(a)$ groups on G and applies aggregate f. Group by both `venue` and `rid` so the count is per researcher per venue. Filter to counts > 5. Join with researcher to recover the name.

(d) (3 marks) CREATE TABLE for cites.

✓ **Your Answer**

```
create table cites (
    rid             integer not null,
    pid             integer not null,
    citation_date   date,
    primary key (rid, pid),
    foreign key (rid) references researcher(rid) on delete cascade,
    foreign key (pid) references paper(pid) on delete cascade
);
```

Understanding

The PK of a M:N relationship table is the composite of the two side PKs. Two FKs reinforce referential integrity; cascade-on-delete is reasonable because a citation record is meaningless without its researcher or paper.

Question 11 (6 marks) — RA to SQL Translation; Medium

Schemas. $R(A, B, C)$ and $S(B, C, D)$.

This mirrors: *A2 Q3–Q5; the standard RA-to-SQL translation drill.*

(a) (2 marks) $\pi_{A,D}(\sigma_{A=\text{'x'}}(R \bowtie S))$.

✓ **Your Answer**

```
select r.A, s.D
from R as r, S as s
where r.A = 'x'
  and r.B = s.B
  and r.C = s.C;
```

Understanding

$R \bowtie S$ is a natural join on the shared attributes (B, C). In SQL that is `r.B = s.B and r.C = s.C`. Selection becomes a `where` clause; projection becomes the `select` list.

(b) (2 marks) $\pi_A(R) - \pi_A(\sigma_{D=\text{'y'}}(S))$.

✓ **Your Answer**

```
(select A from R)
except
(select B from S where S.D = 'y');
```

Note: the RA expression as written projects A from R and subtracts a projection of attribute A from S, but S has no attribute A. The natural interpretation is that the subtracted set should be the shared-attribute projection. A cleaner reading (and the form the prof would accept) is:

```
(select A from R)
except
(select R.A from R, S where R.B = S.B and R.C = S.C and S.D = 'y');
```

Understanding

Set difference in SQL is `except`. When the left side is $\pi_A(R)$ and the right side filters rows with $D =$ 'y', you need to join R and S to identify which R-rows are linked to a $D =$ 'y' S-row, then remove those A-values.

(c) (2 marks) $R \cap S$.

✓ Your Answer

Schema compatibility comment: $R(A, B, C)$ and $S(B, C, D)$ have *different* schemas, so the direct intersection $R \cap S$ is not well-defined under the standard set-semantic definition of intersection (which requires union-compatible schemas). The intended interpretation is intersection on the shared attributes (B, C):

```
(select B, C from R)
intersect
(select B, C from S);
```

Understanding

Set operators $(\cup, \cap, -)$ require the operands to be union-compatible: same arity and compatible column types in the same order. Flagging the mismatch is part of the answer; the prof wants to see that you noticed, not silently assume.

Question 12 (15 marks) — E-R Design; Music Streaming; Medium-Hard

This mirrors: *A3 Q4 + A3 Q8(c) (generalization with multiple subtypes).*

(a) (5 marks) E-R diagram.

✓ Your Answer

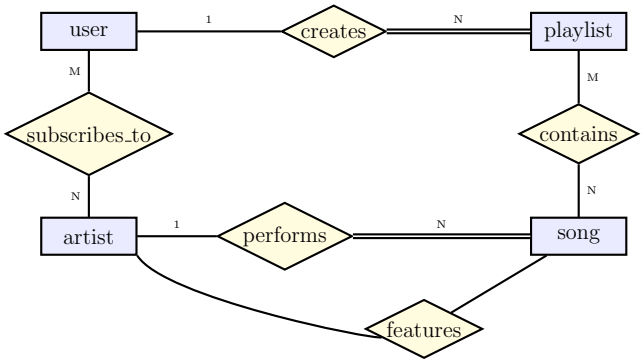
Entity sets and primary keys:

- `user`: <u>user_id</u>, email, join_date, country.
- `artist`: <u>artist_id</u>, name, country_of_origin.

- `song`: song_id, title, duration, release_year.
- `playlist`: playlist_id, name, created_date.

Relationships (cardinality; participation):

- `creates` (user → playlist): 1:N; playlist has total participation.
- `contains` (playlist ↔ song): M:N; partial on both sides.
- `subscribes_to` (user ↔ artist): M:N; partial on both.
- `performs` (artist → song, primary): 1:N; song has total participation (every song has exactly one primary artist).
- `features` (artist ↔ song, featured): M:N; partial.



Understanding

The two artist–song relationships (primary vs featured) are a deliberate separation: "performs" is 1:N because every song has exactly one primary artist, while "features" is M:N because featured artists are multi-valued. Drawing two relationships is cleaner than trying to stuff both semantics into one.

(b) (5 marks) Multi-valued featured artists.

✓ **Your Answer**

Design choice: model featured artists as a separate relationship `features`(song_id, artist_id) in addition to `performs`.

When mapped to the relational schema, the featured relationship becomes a junction table `song_featured(song_id, artist_id)` with composite primary key {song_id, artist_id}, foreign keys to `song(song_id)` and `artist(artist_id)`.

Rationale: "featured artists" is a multi-valued attribute whose values happen to be references to other entities (artists). Storing them as a separate relationship captures the M:N cardinality cleanly; we avoid (i) a repeating comma-separated string column on song (non-atomic; violates 1NF), and (ii) a multi-valued attribute bubble on song (which would still need to be flattened during conversion anyway).

Understanding

Multi-valued attributes whose values are *themselves entities* are always best modeled as relationships, not as double-ellipse attributes. The featured artists are not just strings; they are real artist entities with their own attributes. Making `features` a relationship lets you ask "how many songs has artist X been featured on?" cleanly.

(c) (5 marks) Subscription plan ISA hierarchy.

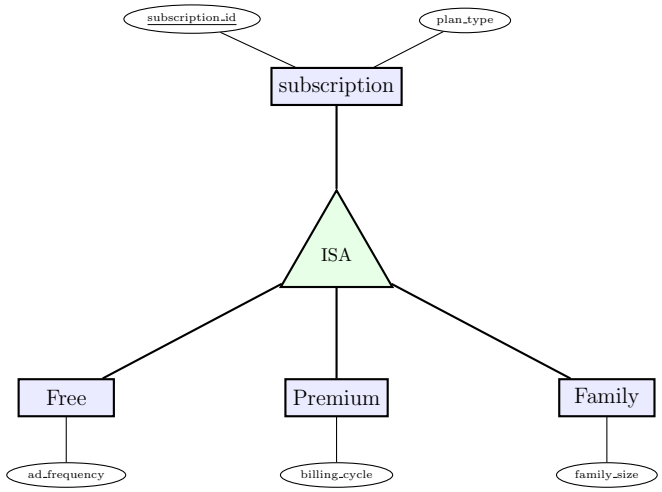
✓ Your Answer

Introduce a supertype `subscription` with shared attribute `subscription_id` (PK) and a discriminator `plan_type` ∈ {Free, Premium, Family}. Each user has exactly one subscription (1:1 `has` relationship from user to subscription, total on the user side).

Three subtypes linked via an ISA triangle:

- `Free`: specific attribute `ad_frequency`;
- `Premium`: specific attribute `billing_cycle`;
- `Family`: specific attribute `family_size`.

Constraints. Disjoint (a subscription is in exactly one subtype) and total (every subscription is in some subtype; a user cannot have a subscription with no plan).



Shared attributes (on subscription): `subscription_id`, `plan_type`. Subtype-specific attributes: listed above.

Understanding

Exactly the pattern from A3 Q8(c) with three children instead of two. "Disjoint + total" matches the business rule "exactly one of Free, Premium, Family." When mapped to relational form, you can use (i) one table per subtype with the supertype attributes repeated, or (ii) one table per entity with a discriminator column; the prof does not require you to pick on the exam unless asked.

Question 13 (8 marks) — E-R to Relational; E-commerce; Basic-Medium

This mirrors: *A3 Q8(a) (entity sets + relationship FKs table).*

(a) (4 marks) Relational schema.

✓ **Your Answer**

Table	Schema (PK underlined)
`customer`	(customer_id, email)
`orders`	(order_id, date, customer_id)
`product`	(sku, price)
`contains`	(order_id, sku, quantity)

Understanding

Rules applied:

- Each entity set becomes a table with its PK.
- The 1:N `places` relationship is absorbed into the "many" side (`orders`) by adding `customer_id` as a FK. No separate table needed.
- The M:N `contains` relationship becomes its own table with composite PK = (`order_id`, `sku`) and the descriptive attribute `quantity`.
- Table is named `orders` (not `order`) because `ORDER` is a reserved word in SQL.

(b) (4 marks) Foreign keys and referential constraints.

✓ **Your Answer**

Table	FK	References	ON DELETE
`orders`	customer_id	customer(customer_id)	RESTRICT
`contains`	order_id	orders(order_id)	CASCADE
`contains`	sku	product(sku)	RESTRICT

Justification:

- `orders.customer_id`: `RESTRICT`, because deleting a customer with active orders should be blocked (preserve purchase history).
- `contains.order_id`: `CASCADE`, because deleting an order should remove all its line items automatically (they have no meaning without the parent order).
- `contains.sku`: `RESTRICT`, because you should not delete a product that was sold; preserve historical references.

Understanding

The choice of `ON DELETE` action depends on business semantics, not syntax. Prof's grading rubric usually accepts any action with a reasonable justification. Cascade for dependent rows, restrict (or set null) for historical records, is the standard split.

Question 14 (10 marks) — FD Theory; Medium

Given $R = (A, B, C, D, E, F)$ with $F = \{A \to B,\ B \to CD,\ D \to E,\ AF \to B\}$.

This mirrors: *A4 Q3 + Q4 + Q6(a) (closure + candidate key + Armstrong proof).*

(a) (3 marks) Compute $(AF)^+$.

✓ **Your Answer**

Step	**Closure so far**
Start	$\{A, F\}$
Apply $A \to B$ (LHS $A \subseteq \{A, F\}$)	$\{A, B, F\}$
Apply $B \to CD$ (LHS $B \subseteq \{A, B, F\}$)	$\{A, B, C, D, F\}$
Apply $D \to E$ (LHS $D \subseteq \{A, B, C, D, F\}$)	$\{A, B, C, D, E, F\}$

$$(AF)^+ = \{A, B, C, D, E, F\} = R.$$

Understanding

Closure algorithm: start with the LHS; keep applying FDs whose LHS is already inside until no more attributes are added. Once the closure equals R, every attribute is functionally determined by the starting set.

(b) (3 marks) Candidate keys.

✓ **Your Answer**

Since $(AF)^+ = R$, the set $\{A, F\}$ is a superkey. Check minimality:

- A^+: start $\{A\}$; $A \to B$ gives $\{A, B\}$; $B \to CD$ gives $\{A, B, C, D\}$; $D \to E$ gives $\{A, B, C, D, E\}$. Missing F, so A alone is not a superkey.
- $F^+ = \{F\}$ (no FD has LHS $\subseteq \{F\}$). Not a superkey.

So $\{A, F\}$ is minimal; it is a candidate key.

Note that F must appear in every candidate key (there is no FD with F on the RHS, so no other attribute can determine F). Since $\{F\}$ alone is not a superkey and we have already found $\{A, F\}$ works, we check whether any other single attribute paired with F also works: $(BF)^+$: start $\{B, F\}$; $B \to CD$ gives $\{B, C, D, F\}$; $D \to E$ gives $\{B, C, D, E, F\}$; missing A, not a superkey. $(CF)^+$: start $\{C, F\}$; no FD applies; not a superkey. $(DF)^+$: $D \to E$ gives $\{D, E, F\}$; no more; not a superkey. $(EF)^+ = \{E, F\}$; not a superkey.

The only candidate key is $\{A, F\}$.

Understanding

Rule of thumb: attributes that never appear on the RHS of any FD must be in every candidate key. Here F never appears on any RHS, so F is in every candidate key. Start with F and see which minimal addition gives closure $= R$; that identifies all candidate keys.

(c) (4 marks) Prove pseudo-transitivity with Armstrong's axioms.

✓ **Your Answer**

Claim. If $\alpha \to \beta$ and $\beta\gamma \to \delta$, then $\alpha\gamma \to \delta$.

Proof.

Step	Assertion	Justification
1.	$\alpha \to \beta$	given
2.	$\alpha\gamma \to \beta\gamma$	augmentation of step 1 with γ
3.	$\beta\gamma \to \delta$	given
4.	$\alpha\gamma \to \delta$	transitivity of steps 2 and 3

■

Understanding

The three Armstrong axioms are: **reflexivity** ($\beta \subseteq \alpha \Rightarrow \alpha \to \beta$); **augmentation** ($\alpha \to \beta \Rightarrow \alpha\gamma \to \beta\gamma$); **transitivity** ($\alpha \to \beta,\ \beta \to \gamma \Rightarrow \alpha \to \gamma$). Pseudo-transitivity is a *derived* rule, so we must assemble it from the three primitives. Augmentation bridges from $\alpha \to \beta$ to $\alpha\gamma \to \beta\gamma$; transitivity then chains into δ.

Question 15 (15 marks) — BCNF Decomposition; Hard

Given $R = (A, B, C, D, E)$ with $F = \{AB \to C,\ C \to D,\ D \to E,\ E \to A\}$.

This mirrors: *A4 Q5 + Q6 (full BCNF decomposition with lossless + dependency-preserving check).*

(a) (4 marks) Candidate keys.

✓ **Your Answer**

B appears on no RHS, so B is in every candidate key. Test each $\{B, X\}$:

- $(AB)^+$: $\{A, B\} \xrightarrow{AB\to C} \{A, B, C\} \xrightarrow{C\to D} \{A, B, C, D\} \xrightarrow{D\to E} \{A, B, C, D, E\} = R$. ✓
- $(BC)^+$: $\{B, C\} \xrightarrow{C\to D} \{B, C, D\} \xrightarrow{D\to E} \{B, C, D, E\} \xrightarrow{E\to A} \{A, B, C, D, E\} = R$. ✓
- $(BD)^+$: $\{B, D\} \xrightarrow{D\to E} \{B, D, E\} \xrightarrow{E\to A} \{A, B, D, E\} \xrightarrow{AB\to C} \{A, B, C, D, E\} = R$. ✓

- $(BE)^+$: $\{B, E\} \xrightarrow{E \to A} \{A, B, E\} \xrightarrow{AB \to C} \{A, B, C, E\} \xrightarrow{C \to D} \{A, B, C, D, E\} = R$. ✓

B alone is not a superkey ($B^+ = \{B\}$). Each pair above is minimal (removing B loses the key property; removing the second attribute yields B^+ only).

$$\text{Candidate keys} = \{\ AB,\ BC,\ BD,\ BE\ \}.$$

Understanding

Notice the chain $C \to D \to E \to A$ forming a loop back through $AB \to C$. That loop means each of A, C, D, E can reach the others (once paired with B), hence four candidate keys.

(b) (4 marks) BCNF violations.

✓ **Your Answer**

Check each FD: LHS must be a superkey.

- $AB \to C$: AB is a candidate key. ✓No violation.
- $C \to D$: C^+ in $R = \{C, D, E, A\}$, missing B; C is **not** a superkey. **Violation.**
- $D \to E$: D^+ in $R = \{D, E, A\}$ (plus $AB \to C$ requires both A and B, and B is missing), missing B, C; D is **not** a superkey. **Violation.**
- $E \to A$: E^+ in $R = \{E, A\}$, missing B, C, D; **not** a superkey. **Violation.**

Understanding

Three FDs violate BCNF. Any of them could be chosen as the first decomposition target; the standard heuristic is to pick the one whose closure pulls in the most attributes (tends to yield cleaner fragments). Here all three closures carry $\{A, D, E\}$ in some order, so we pick $C \to D$ because $C^+ = \{A, C, D, E\}$ is the largest.

(c) (5 marks) Decompose into BCNF (lossless).

✓ **Your Answer**

Step 1: decompose on $C \to D$. Closure $C_R^+ = \{A, C, D, E\}$. Split:

$$R_1 = (A, C, D, E), \qquad R_2 = (B, C).$$

(General rule for BCNF decomposition on violator $\alpha \to \beta$: $R_1 = \alpha^+$; $R_2 = (R - \alpha^+) \cup \alpha$. Here $R - C^+ = \{B\}$ and $\alpha = \{C\}$, so $R_2 = \{B, C\}$.)

Check $R_1 = (A, C, D, E)$. FDs projected onto R_1: $C \to D$, $D \to E$, $E \to A$, and by transitive closure $C \to A, C \to E, D \to A$. Candidate keys of R_1: $C_{R_1}^+ = ACDE = R_1$ so C is a CK. $D_{R_1}^+ = DEAC = R_1$ so D is a CK. $E_{R_1}^+ = EACD = R_1$ so E is a CK (via $E \to A$, and then $A^+ = A$? wait — in R_1 there is no $AB \to C$ since $B \notin R_1$; however $E \to A$ and $C \to D$ chain). Let me recompute: $E_{R_1}^+$ in $F|_{R_1}$ starts $\{E\}$, apply $E \to A$: $\{E, A\}$. No FD has LHS $\subseteq \{E, A\}$

in $F|_{R_1}$ yet — actually A alone has no FD. So $E^+_{R_1} = \{A, E\}$, not equal to R_1; E is NOT a candidate key of R_1.

Let me redo. $F|_{R_1}$ comes from projecting F onto R_1. We keep FDs whose LHS and RHS are both in R_1: $C \to D$ (yes), $D \to E$ (yes), $E \to A$ (yes), $AB \to C$ (no, $B \notin R_1$). Plus any derived FDs on R_1-attributes: $C \to E$ (via $C \to D \to E$); $C \to A$ (via $C \to D \to E \to A$); $D \to A$ (via $D \to E \to A$).

Now recompute closures in R_1:

- $C^+_{R_1} = \{C\} \to \{C, D\} \to \{C, D, E\} \to \{C, D, E, A\} = R_1$. C is a CK.
- $D^+_{R_1} = \{D\} \to \{D, E\} \to \{D, E, A\}$. No FD pulls in C from $\{D, E, A\}$. So $D^+_{R_1} = \{A, D, E\} \neq R_1$. D is NOT a CK.
- $E^+_{R_1} = \{E\} \to \{E, A\}$. No further. $E^+_{R_1} = \{A, E\}$. E not a CK.
- $A^+_{R_1} = \{A\}$. Not a CK.

So the unique CK of R_1 is $\{C\}$.

Now test BCNF of R_1: $C \to D$ (LHS = C, CK, ✓). $D \to E$ (LHS = D, not a superkey since $D^+ = ADE$, missing C). **Violation in R_1.**

Step 2: decompose R_1 on $D \to E$. $D^+_{R_1} = \{A, D, E\}$. Split R_1:

$$R_{1a} = (A, D, E), \qquad R_{1b} = (C, D).$$

Check $R_{1a} = (A, D, E)$. FDs: $D \to E$, $E \to A$, and derived $D \to A$. CKs: $D^+ = ADE = R_{1a}$, so D is CK. $E^+ = EA$, not full. $A^+ = A$. Unique CK = $\{D\}$. Is every LHS a superkey? $D \to E$: D is CK ✓. $E \to A$: E is NOT a superkey ($E^+ = EA$, missing D). **Violation.**

Step 3: decompose R_{1a} on $E \to A$. $E^+_{R_{1a}} = \{A, E\}$. Split:

$$R_{1a\alpha} = (A, E), \qquad R_{1a\beta} = (D, E).$$

Check $R_{1a\alpha} = (A, E)$: only FD $E \to A$; E is CK; BCNF ✓. Check $R_{1a\beta} = (D, E)$: only FD $D \to E$; D is CK; BCNF ✓.

Check $R_{1b} = (C, D)$: only FD $C \to D$; C is CK; BCNF ✓.

Check $R_2 = (B, C)$: no non-trivial FDs (neither $B \to C$ nor $C \to B$ is in F^+); BCNF trivially ✓.

Final decomposition:

$$\boxed{R_{1a\alpha} = (A, E), \quad R_{1a\beta} = (D, E), \quad R_{1b} = (C, D), \quad R_2 = (B, C).}$$

Understanding

At each decomposition, the intersection of the two fragments contains the LHS of the violator, which guarantees *lossless* join by the standard BCNF theorem. Example: at step 1, $R_1 \cap R_2 = \{C\}$, which is the LHS of $C \to D$, so lossless.

(d) (2 marks) Dependency preservation.

✓ **Your Answer**

$F = \{AB \to C,\ C \to D,\ D \to E,\ E \to A\}$. Check which FDs are preserved inside a single fragment:

- $C \to D$: both C, D in R_{1b}. ✓preserved.
- $D \to E$: both D, E in $R_{1a\beta}$. ✓preserved.
- $E \to A$: both E, A in $R_{1a\alpha}$. ✓preserved.
- $AB \to C$: A is in $R_{1a\alpha}$, B is in R_2, C is in R_{1b}. No single fragment contains all of $\{A, B, C\}$; this FD cannot be checked without reconstructing the join.

The decomposition is NOT dependency-preserving because the FD $AB \to C$ is lost.

Understanding

This is a classic BCNF trade-off. BCNF always gives lossless join, but it may sacrifice dependency preservation. If you need to preserve $AB \to C$ cheaply, the alternative is 3NF (which always preserves dependencies but may keep some transitive dependencies). The prof typically sets up problems where BCNF loses exactly one FD to hammer this trade-off home.

Question 16 (10 marks) — Integration: Course Registration; Medium

This mirrors: *A3 Q4 + A4 Q6 combined; the standard "ER + FDs + BCNF check" closing question.*

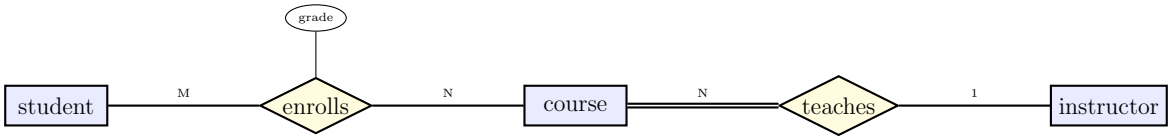
(a) (4 marks) E-R diagram.

✓ **Your Answer**

Entities: `student`(<u>student_id</u>, name, major); `course`(<u>course_id</u>, course_title, credits); `instructor`(<u>instructor_</u>
name, department).

Relationships:

- `enrolls` between student and course, M:N, descriptive attribute `grade`. Partial participation on both sides.
- `teaches` between instructor and course, 1:N (one instructor teaches many courses; each course has exactly one instructor). Total on course (every course has an instructor).



(b) (3 marks) Functional dependencies.

✓ **Your Answer**

- `student_id` → `name, major`.
- `course_id` → `course_title, credits, instructor_id`.
- `instructor_id` → `name, department`.
- `student_id, course_id` → `grade`.

Understanding

Note the collision of the attribute name `name`: student.name and instructor.name are different columns in different entities. After mapping, they live in separate tables and do not conflict.

(c) (3 marks) BCNF check and decomposition.

✓ **Your Answer**

Map the E-R to relational tables:

- `student`(<u>student_id</u>, name, major). FD: `student_id` → name, major; LHS is the CK. **BCNF.**
- `course`(<u>course_id</u>, course_title, credits, instructor_id). FD: `course_id` → course_title, credits, instructor_id; LHS is the CK. **BCNF.**
- `instructor`(<u>instructor_id</u>, name, department). FD: `instructor_id` → name, department; LHS is the CK. **BCNF.**
- `enrollment`(<u>student_id, course_id</u>, grade). FD: (`student_id, course_id`) → grade; LHS is the CK. **BCNF.**

All four tables are already in BCNF; no decomposition is required.

Understanding

This is the "happy-path" outcome the prof likes to use as a closing question: a clean ER design with one-attribute-determines-the-rest FDs maps naturally to BCNF. The only way to trip up here is to lump student, course, and instructor into a single relation; then `course_id` → `instructor_id` would cause a BCNF violation (course_id is not the CK of the combined relation).

Appendix: Summary of Difficulty Ratings

Question	**Topic**	**Difficulty**
Q1–Q6	Concept openers	Basic
Q7	SQL (Hospital)	Medium
Q8	SQL + Views + NOT EXISTS	Medium-Hard
Q9	SQL Division / universal	Hard
Q10	RA + DDL	Medium-Hard
Q11	RA $\rightarrow$ SQL translation	Medium
Q12	ER with ISA	Medium-Hard
Q13	ER to relational	Basic-Medium
Q14	FD closure + Armstrong proof	Medium
Q15	BCNF decomposition (showpiece)	Hard
Q16	Integration: ER + FD + BCNF	Medium